

Subtracting Across a Zero

Grades 2–3 Number and Operations in Base Ten

Regrouping across one zero, regrouping across two zeros, estimating the difference, and checking by adding

How to trade when there is a zero. You cannot take ones from a zero, and you cannot take tens from a zero. Go left until you find a digit you *can* trade, then trade back one place at a time.

Worked model: $305 - 168$

1. Look at the tens.

There are 0 tens, so the tens cannot give the ones a group. Go left.

2. Trade a hundred.

305 is 3 hundreds, 0 tens, 5 ones. Trade 1 hundred for 10 tens: 2 hundreds, 10 tens, 5 ones.

3. Trade a ten.

Now the tens can trade: 2 hundreds, 9 tens, 15 ones. Subtract: $15 - 8 = 7$, $9 - 6 = 3$, $2 - 1 = 1$.

So $305 - 168 = 137$. Check it: $137 + 168 = 305$.

Directions. Subtract. Show every trade you make above the digits. Write your final answer on the answer line.

1. Subtract.

$$\begin{array}{r} 302 \\ - 147 \\ \hline \end{array}$$

Answer: _____

2. Subtract.

$$\begin{array}{r} 605 \\ - 238 \\ \hline \end{array}$$

Answer: _____

3. Subtract. Both the ones and the tens will need a trade.

$$\begin{array}{r} 704 \\ - 526 \\ \hline \end{array}$$

Answer: _____

4. **Estimate first.** Round *each* number to the nearest hundred, then subtract the rounded numbers. Do not find the exact answer here — just the estimate.

$$608 - 293 \approx \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

Answer: _____

5. Subtract. This one has *two* zeros to travel across.

$$\begin{array}{r} 500 \\ - 236 \\ \hline \end{array}$$

Answer: _____

6. Kim subtracted $502 - 138$ and wrote 364. Check her answer by adding her answer back to the number she subtracted: find $364 + 138$. Then write what your check tells you about Kim's answer.

Answer: _____

7. Subtract.

$$\begin{array}{r} 800 \\ - 345 \\ \hline \end{array}$$

Answer: _____

- 8. Estimate.** Round *each* number to the nearest ten, then subtract the rounded numbers.

$$702 - 268 \approx \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

Answer:

- 9. Find and fix the mistake.** Rico worked $903 - 458$ like this: “ $8 - 3 = 5$, $5 - 0 = 5$, $9 - 4 = 5$, so the answer is 555.” Explain in one sentence what Rico did wrong, then work the problem correctly.

$$\begin{array}{r} 903 \\ - 458 \\ \hline \end{array}$$

Answer:

- 10.** Subtract. Start at 1000 and trade all the way from the thousands.

$$\begin{array}{r} 1000 \\ - 487 \\ \hline \end{array}$$

Answer: